MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS NICKEL-ALUMINUM, TYPE I, CLASS 2, COMPOSITION H

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Ni	Al	Others (1)	
		Each	Total
Bal	4.0 5.5	-	2.5 max.

(1) Present as volatile phenolic binder.

Particle Size/Distribution

+140 Mesh (+106 µm)	0.5 max.
+170 Mesh (+90 µm)	5.0 max.
+325 Mesh (+45 µm)	90.0 min.

Metallographic Inspection

When inspected metallographically, powder particles shall be capable of exhibiting a spherical, nickel core essentially surrounded by small aluminum particles.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:32Z
Check the source to verify that this is the current version before use.

Sprayed Coating Properties

Bond Strength	34.0 MPa (5000 psi) min.
Hardness	Rockwell 15T, 82 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 15% max. Unmelted Particles - 2% max.

Deposit Efficiency and Deposition Rate

65% min for N_2/H_2 @ 6.8 lb/hr
60% min for A/H_2 @ 5.4 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

General Electric Company	B50TF56, Class A
Pratt & Whitney Aircraft	PWA 1337
Rolls-Royce, Ltd.	MSRR 9507/5
Allison Gas Turbine	EMS 56757C

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/10)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:32Z
Check the source to verify that this is the current version before use.